An infinite system has no maximum in an HF-ordering and every element lies below a product

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Editorial note. The counted statement is the author's environment at source0.tex lines 829-834 and its proof environment opens at line 835; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

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Editorial scope. One original result of the pinned author manuscript, counted once: no maximum in an HF-ordering of an infinite system, and products above any prescribed element (author label tree2).

Allowed standard prerequisites (editorial).

- HF-ordering of an infinite system. A linear ordering in which each element has at most one pair of smaller elements forming a block with it.. It is the ordering under consideration in both items.
- Block. A block is a triple of elements closed under the partial operation of the Steiner triple system.. It is what the maximality argument counts.

- Infinite Steiner triple system. The system has more than three elements, so each element lies in several blocks.. It is what makes the first item work.

Lemma 0.1 1. An HF-ordering of an infinite STS does not have a maximum.

2. If $<$ is an HF-ordering of an infinite Steiner triple system M, then for every $a \in M$, for every $b \geq a$, there are $b_1, b_2 > b$ such that $a = b_1 \cdot b_2$.

Proof. 1. Assume $a \in M$ is a maximum. Since $|M| > 3$, the element a belongs to two different blocks, $\{a, a_1, a \cdot a_1\}$ and $\{a, a_2, a \cdot a_2\}$. By maximality, $a > a_1, a_1 \cdot a, a_2, a_2 \cdot a$, in contradiction with $<$ being an HF-ordering.

2. Assume not, and take $b \geq a$ such that if $a = a_1 \cdot a_2$, then $a_1 \leq b$ or $a_2 \leq b$. Since M is infinite, by part 1 we can choose $b_1 > b_2 > b$ such that $b_1 \cdot b_2 < b_1$. Note that $\{b_1, a, a \cdot b_1\} \neq \{b_1, b_2, b_1 \cdot b_2\}$, since otherwise $a = b_1 \cdot b_2$ and $b_1, b_2 > b$. If $b_1 > a \cdot b_1$, then the previous blocks contradict that $<$ is an HF-ordering. Hence, $b_1 < a \cdot b_1$. In this case, $a = b_1 \cdot (a \cdot b_1)$ with $b_1, a \cdot b_1 > b$, a contradiction with the choice of b. □